MACHINE LEARNING PROJECT

Car Price Prediction Using Machine Learning

Predicting used-car prices from vehicle features and selecting the best regression model.



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THE PROJECT

The project turns car details into a price estimate

Predict a used car’s selling price from information such as brand, model, year, mileage, fuel type, engine, and safety features.

Why it matters

A data-based estimate can support faster and more consistent pricing decisions.

Approach

Clean the data, study price patterns, train several regression models, and compare their results.



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DATASET

18,924

cleaned car records

Started with 19,237 rows and 18 columns.

The dataset contains 18,924 cleaned car records

313

duplicate rows removed

Numeric cleanup

Converted mileage into a number and cleaned levy and engine-volume values.

Missing values

Filled missing levy values with the median.

Outlier rule

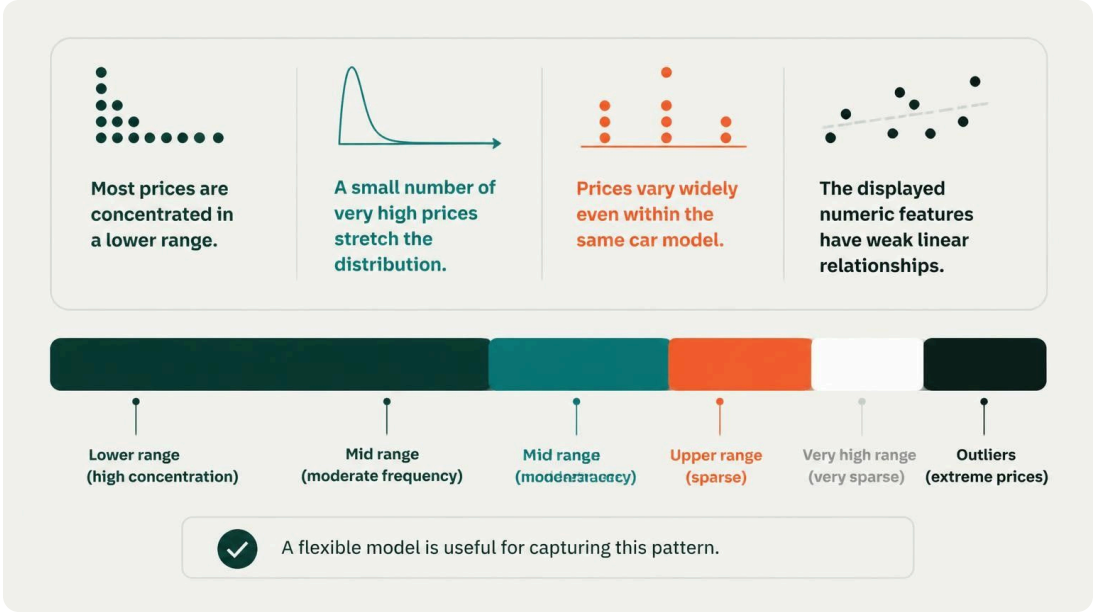
Handled unusual values before modeling.

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EXPLORATORY DATA ANALYSIS

Price data is highly uneven, so cleaning matters

- Most prices are concentrated in a lower range, while a small number of very high prices stretch the distribution.
- Prices can vary widely even within the same car model.
- The correlation check shows weak linear relationships with the displayed numeric features, so a flexible model is useful.



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MODELING WORKFLOW

The workflow combines preparation, training, and testing

01

Separate

Separate the target price from the input features.

02

Prepare

Encode text categories such as model, fuel type, and color; scale numeric features where needed.

03

Test

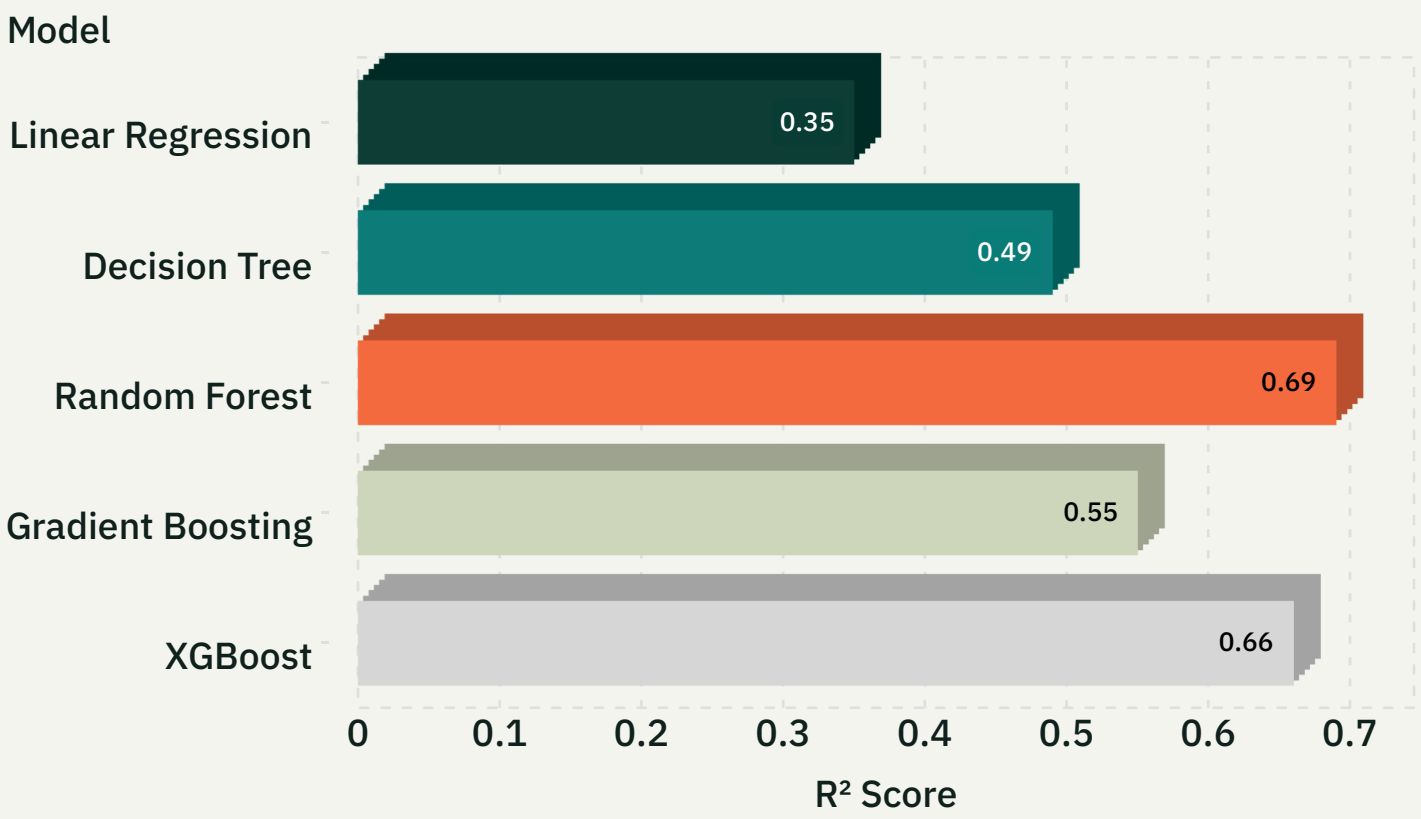
Split the data into training and test sets, then evaluate predictions on unseen data.

The test set checks how the models perform on data they have not seen.

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MODEL COMPARISON

Random Forest gives the strongest test result



Test results

0.69

Random Forest R^2

3,735.74

MAE

6,233.29

RMSE

Random Forest leads the five tested models.

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DEPLOYMENT DECISION

Random Forest is the selected model for deployment

150

trees selected through hyperparameter tuning

25

maximum depth

≈ 0.70

tuned cross-validation R^2 score



Ready for new listings

Key lesson: careful cleaning and model comparison are as important as the algorithm itself.

Next step: test the model on new listings and monitor prediction quality over time.

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